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Chris Smith

I am a physical climate scientist with more than 10 years' experience, working at the interface of Earth System modelling and scenario projections. Presently, I am best known as the developer of the FaIR reduced-complexity climate model that was used extensively in the 2018 IPCC Special Report on Global Warming of 1.5°C, the IPCC Sixth Assessment (AR6) Working Group I (WG1) and Working Group III (WG3) reports, and several high-impact academic and policy outputs. I was a key contributor to the Reduced Model Intercomparison Project (RCMIP) that benchmarked the performance of emulators for use in the IPCC AR6. I am also a central part of the team that annually updates IPCC WG1 physical climate assessments based on new data (the Indicators of Global Climate Change project). I have been selected as a Coordinating Lead Author for the IPCC Seventh Assessment Report in WG1 Chapter 5 (Scenarios and Projected Future Global Temperatures).
The other major strand of my research is the Earth's energy budget and radiative forcing. I led and contributed to several important papers analysing the radiative forcing from CMIP6 era models, forming a critical component of the radiative forcing assessments in IPCC AR6 WG1. I am heavily involved in the design and delivery of the Coupled Model Intercomparison Project Phase 7 (CMIP7), having served as a member of the Strategic Ensemble Design Task Team, a current co-chair of the Radiative Forcing Model Intercomparison Project (RFMIP), and an advisory board member to the Scenario Model Intercomparison Project (ScenarioMIP).

Career experience

2026– **Senior Research Scholar**, *Energy Climate and Environment Program, International Institute for Applied Systems Analysis*, Laxenburg, Austria

2024–2025 **Senior Scientist**, *Department of Water & Climate, Vrije Universiteit Brussel*, Brussels, Belgium

2023–2024 **Climate Mitigation Expert Scientist**, *UK Met Office*

2015–2024 **Senior Research Fellow**, *School of Earth and Environment, University of Leeds*

Prizes and recognition

2025 Highlight talk, EGU 2025 *Updated IPCC emissions scenarios no longer limit warming to 1.5°C*

2025 Clarivate Highly Cited Researcher

2024–2025 Stanford/Elsevier Top 2% Scientists 2024

2023 Winner of the Societal Impact Award, UK Natural Environment Research Council

2018– Coordinating Lead Author, Lead Author, Chapter Scientist and Contributing Author in several roles across three reports of the Intergovernmental Panel on Climate Change (Seventh Assessment Report (AR7) Working Group 1. Sixth Assessment Report (AR6) Working Group 1, AR6 Working Group 3, and Special Report on Global Warming of 1.5°C)

2016 Outstanding reviewer contribution award, Renewable Energy journal

Research funding

2026–2030 Task leader, FUTURA, Horizon Europe (consortium €30m)

2026–2028 International co-investigator, SPECTRE (led by U. Leeds), UK Research and Innovation (consortium £1m)

2026–2027 Co-investigator, understanding Earth's Energy Budget (led by U. Illinois), SPARK Climate

2024–2028 Principal Investigator, HYWAY, Horizon Europe (consortium €4.6m)

2024–2027 Work Package Leader, Climate Scenarios Compass Initiative, Bezos Earth Fund (consortium €2.2m)

2024–2026 Co-Investigator, MAGICA (led by U. Leeds), UK Natural Environment Research Council (consortium £1m)

2024–2026 Co-Investigator, Contrail assessment of future aircraft and propulsion architectures (led by U. Southampton), UK Natural Environment Research Council (consortium £1m)

2023–2026 Principal Investigator & Work Package leader, WorldTrans, Horizon Europe (consortium €5m)

2023–2026 Steering committee member, TRIFECTA, Norwegian Research Council

2020–2024 Individual Fellowship, NERC-IIASA Collaborative Research Fellowship, Natural Environment Research Council (£390k)

2017–2019 Co-Investigator, CAMS74, Copernicus Atmosphere Monitoring Service radiative forcing products (£101k)

Student and postdoctoral supervision

2023– Postdoctoral supervisor, Chris Wells, University of Leeds (WorldTrans)

2023– Lead PhD supervisor, Alejandro Romero Prieto, University of Leeds (PANORAMA Doctoral Training Programme, supported by Met Office CASE partner) *Efficient models for climate mitigation and adaptation*

2023–2024 MRes supervisor, Yongyao Liang, University of Leeds *Attribution of 2023's record temperature*

2019–2024 Seven MSc and undergraduate dissertation students at U. Leeds, two co-authored publications [32, 79]

Leadership and professional activities

2025 Expert reviewer for Chancery Lane project for legal definitions of climate change

2024–2025 Scenarios Forum 2025 (hosted by University of Leeds); part of winning bid team, and member of Scientific Steering Council and organizational committee

2023–2026 External reviewer for individual and team research funding proposals in Switzerland, Canada and the US

2023– Co-chair, Radiative Forcing Model Intercomparison Project (RFMIP) for CMIP7

2023– Committee member, CACTI (Composition Air-quality Climate inTeractions Initiative)

2023–2024 External examiner for two PhD thesis defences in Canada and France

2022–2024 Task Team Member, CMIP7 Strategic Ensemble Design Task Team

2021– External consultant on applications of reduced complexity climate models for high-profile clients including national governments

2021 Panel member at COP26 IPCC side event, Glasgow, UK

2020–2025 Convenor of European Geosciences Union sessions on climate modelling, climate emulation, economics, atmospheric chemistry, and air pollution

2019 Scientific expert, British High Commission Malaysia Communicating Climate Science initiative, Malaysia

Media and outreach

2026 Equinox workshop: Belspo, Brussels, Belgium *10 Years after the Paris Agreement*

2025 Press conference: "Hot takes & policy quakes: When geoscience meets social science", European Geosciences Union, Vienna, Austria *On the feasibility of current scenarios to limit warming to 1.5°C*

2024 Live national radio interview: BBC Radio Five Live *On the climate ambitions of the two main political parties in the UK*

2024, 2026 Press conference: United Nations Framework Convention on Climate Change (UNFCCC) SBSTA60/62, Bonn, Germany *Indicators of Global Climate Change: Annual updates to IPCC climate system assessments*

2024 TV interview: The Nation Kenya *Solutions for climate change mitigation*

2024 YouTube interview: British Embassy in Vienna *Climate science with a message*

2023 Live national radio interview: BBC Radio 4 Today Programme *Northern Hemisphere 2023 heatwave*

2022 Live TV interview: TVP Poland *Climate impacts of NordStream gas pipeline leak*

2022 TV interview: BBC Two Newsnight *UK summer heatwave*

2018 Live TV interview: Sky News *IPCC Special Report on 1.5°C*

2018– Several guest posts on Carbon Brief and The Conversation climate-focused online news sources

2018– Several interviews on local radio and for newspapers in the UK, US, Germany, Austria, Australia, France, Norway, Sweden, Canada and Belgium. Examples include BBC, CNN, New York Times, Washington Post, The Guardian

Invited conference talks, workshops and panels

2026 World Young Scientist Summit, Wenzhou, China

2026 Climate Negotiation and Fair Climate Policy, Lorentz Center, Leiden, Netherlands

2025 Belgian Ministry of Health: 10 years since the Paris Agreement, Brussels, Belgium

2025 ESA Living Planet Symposium panel discussion, Vienna, Austria

2024 Climate economics conference, Zürich, Switzerland

2024 Econometric Models of Climate Change, Cambridge, UK

2024 European Space Agency TRUTHS mission launch, Didcot, UK

2024– Transient Climate Response to Emissions workshops (Bristol, UK; Vienna, Austria)

2023 Workshop on new modelling framework for climate mitigation, Paris, France

2023 Gordon Research Conference on Radiation and Climate, Maine, USA

2023 Climate emulator workshop, Vienna, Austria

2022 Global Warming Levels workshop, Reading, UK

2022 Institute for Mathematical and Statistical Innovation, Chicago, USA

2022 Platform for Advanced Scientific Computing Conference, Basel, Switzerland

2020 Institute of Physics, London, UK

2018 American Geophysical Union Fall Meeting, Washington DC, USA

2017– PDRMIP, TriMIPathlon & CACTI workshops (Imperial College, UK; IPSL, France; Princeton, US; Kiel, Germany; and online)

Invited institutional seminars

2025 ETH Zürich, Switzerland
2025 University of Oslo, Norway
2025 Yonsei University, Seoul, South Korea
2024 Vrije Universiteit Brussel, Belgium
2024 Beijing Institute of Technology, China
2024 University of Oslo, Norway
2022 University of Zürich, Switzerland
2022 BOKU, Vienna, Austria
2021 University of Cambridge, UK
2020 Energy Meteorology Group, University of Reading, UK
2019 CICERO, Oslo, Norway
2019 Imperial College, London, UK
2019 University of East Anglia, Norwich, UK
2018 Geophysical Fluid Dynamics Laboratory (GFDL), Princeton, New Jersey, USA
2018 University of Manchester, UK

Visiting scientist positions

2025 Delft University of Technology, Delft, Netherlands
2025– Met Office Hadley Centre, Exeter, UK
2024– Priestley Centre for Climate Futures, University of Leeds, UK
2024 CICERO, Oslo, Norway
2020–2025 IIASA, Laxenburg, Austria

Teaching experience

2025– Bergen Summer Research School, Bergen, Norway
2025 Introduction to Physical Climate Change, UCLouvain, Belgium
2018–2024 National Centre for Atmospheric Science PhD Introduction *one-week intensive course for new PhD students from around the UK*. I delivered a lecture and practical session using a simple climate model demonstrating radiative forcing and the Earth's climate response.
2018–2021 Climate Change & Environmental Policy, University of Leeds, UK *One-semester (22 teaching hours) course on the physical climate system aimed at social scientists.* I was module leader in 2020–21, and adapted the course substantially to maintain student interest and engagement due to shifting teaching activities online during the Covid-19 pandemic. Nominated by my students for an Inspiring Teaching Award in 2019.
2016–2021 ad-hoc lecturing in the School of Earth & Environment at University of Leeds, UK *climate-related undergraduate courses.* Topics including climate mitigation scenarios, radiative forcing, the carbon cycle, geoengineering and sustainable food systems.
2013–2015 Module demonstrator, School of Mechanical Engineering, University of Leeds, UK *Thermofluids laboratory practicals*

Code and software development

Proficient developer using collaborative software best practices, releasing version-controlled open-source software on the Python Package Index and Anaconda (fair and climateforcing packages). Experienced user of python, R, MATLAB, FORTRAN, bash, HTML, Git, GitHub, subversion, parallel computing, LaTeX and Unix systems. Experience in handling and processing large climate and observational datasets (netcdf4, hdf5). Experienced user of radiative transfer (SOCRATES, libRadtran, RFM) and complex climate (UKESM and HadGEM family) models, including code development.

Education, qualifications, and previous career history

2011–2015 **Integrated MSc and PhD in Low Carbon Technologies**, *School of Chemical and Process Engineering, University of Leeds*, UK

2008–2011 **International tax**, *Deloitte*, Nottingham, UK

2004–2008 **Bachelor and Master of Mathematics (MMath)**, *University of Nottingham*, UK

Publications

Reports

Intergovernmental Panel on Climate Change (IPCC)

Coordinating Lead Author

- **Seventh Assessment Report (AR7) Working Group 1 Chapter 5 (2025–2028) *Scenarios and Projected Future Global Temperatures***
- Sixth Assessment Report (AR6) Working Group 1 Annex III, *Tables of Historical and Projected Well-mixed Greenhouse Gas Mixing Ratios and Effective Radiative Forcing of All Climate Forcers* (Dentener, Hall, **Smith** et al., 2021)

Lead Author

- AR6 Working Group 1 Chapter 7 Supplementary Material *The Earth's Energy Budget, Climate Feedbacks and Climate Sensitivity* (**Smith** et al. 2021)

Chapter Scientist

- AR6 Working Group 1 Chapter 7 (2021)
- Special Report on 1.5°C Chapter 2 (2018)

Contributing Author

- AR6 Working Group 3 Chapter 3 (2022)
- AR6 Working Group 1 Summary for Policymakers (2021)
- AR6 Working Group 1 Chapter 2 (2021)
- AR6 Working Group 1 Chapter 4 (2021)
- AR6 Working Group 1 Chapter 6 (2021)
- AR6 Working Group 1 Chapter 7 (2021)
- Special Report on 1.5°C Chapter 1 (2018)

Other reports

- Targets for effective climate mitigation governance in the UK. Climate Evidence Unit, Leeds, 2024.
- What will climate change cost the UK? A study of climate risks, impacts and mitigation for the net-zero transition. London School of Economics, London, 2022.
- ZERO IN ON series, produced under the CONSTRAIN Horizon Europe project 2019–2021 (vol. 3 | vol. 1)

Peer-reviewed journal articles

2026

123. Al-Yaari A, **Smith C**, Yao Y, Aas KS, Leung LR, Lo M-H, McDermid S, Nazarenko L, Pokhrel Y, Satoh Y, Yokohata T, Thiery W, 2026. Past and future projected radiative effects from irrigation. Earth System Dynamics, 17(6), 1629–1647. doi:10.5194/esd-17-1629-2026

122. Partanen A-I, Ekholm T, Seppälä J, Weaver S, Palosuo T, Grassi G, **Smith C**, Kulovesi K, Laine H, Ollikainen M, 2026. A new approach to determine national greenhouse gas budgets. Environmental Research Letters, 21(16), 164008. doi:10.1088/1748-9326/ae9628
121. Deidda C, Khanna AA, Schade W, **Smith C**, Thiery W, 2026. Europe's transport infrastructure is not ready to face climate change. Natural Hazards and Earth System Sciences, 26(7), 3345–3394. doi:10.5194/nhess-26-3345-2026
120. **Smith C**, Ramme L, Wells CD, Gjermundsen A, Li H, Ilyina T, Muralidhar A, Bourgeois T, Schwinger J, Romero-Prieto A, Li C, Mauritzen C, 2026. Overshoot and (ir)reversibility to 2300 in two CO2-emissions driven Earth System models. Earth System Dynamics, 17(3), 893–911. doi:10.5194/esd-17-893-2026
119. Rising J, Khurana R, Dietz S, Dumas M, Kikstra J, Lenton TM, Linsenmeier M, **Smith C**, Ward B, 2026. Comprehensive national climate damage assessments framework applied to the UK. Nature Climate Change, 16(7), 790–795. doi:10.1038/s41558-026-02665-2
118. Forster PM, Walsh T, **Smith C**, Lamb WF, Lamboll R, Cassou C, Hauser M, Hausfather Z, Lee J-Y, Palmer MD, von Schuckmann K, Slangen ABA, Szopa S, Trewin B, Yun J, Gillett NP, Jenkins S, Matthews HD, Raghavan K, Ribes A, Rogelj J, Rosen D, Zhang X, Allen M, Andrew RM, Atkinson C, Betts RA, Bombelli A, Burgess SN, Cheng L, Claxton HE, Friedlingstein P, Frölicher TL, Domingues CM, Gasser T, Gregory CH, Hoesly RM, Huppmann D, Ishii M, Kadow C, Karwat A, Kennedy J, Killick RE, Kovilakam MVM, Krummel PB, Lan X, Lamarque J-F, Liné A, Martín-Míguez B, Monselesan DP, Morice C, Mühle J, Mussak P, Peters GP, Pirani A, Pongratz J, Rigby M, Rohde R, Savita A, Seneviratne SI, Smith SJ, Taha G, Tassone C, Thorne P, Wells C, Western LM, van der Werf GR, Wijffels SE, Zecchetto M, Zhong J, Zhang X-Y, Masson-Delmotte V, Zhai P, 2026. Indicators of Global Climate Change 2025: annual update of key indicators of the state of the climate system and human influence. Earth System Science Data, 18(6), 3889–3933. doi:10.5194/essd-18-3889-2026
117. Kramer RJ, **Smith C**, Andrews T, 2026. The Radiative Forcing Model Intercomparison Project (RFMIP2.0) for CMIP7. Geoscientific Model Development, 19(10), 4447–4466. doi:10.5194/gmd-19-4447-2026
116. Lamb WF, Andrew RM, Jones M, Nicholls Z, Peters GP, **Smith C**, Saunois M, Grassi G, Pongratz J, Smith SJ, Tubiello FN, Crippa M, Gidden M, Friedlingstein P, Minx J, Forster PM, 2026. Differences in anthropogenic greenhouse gas emissions estimates explained. Earth System Science Data, 18(4), 2549–2572. doi:10.5194/essd-18-2549-2026
115. Van Vuuren DP, O'Neill BC, Tebaldi C, Sanderson BM, Chini LP, Friedlingstein P, Hasegawa T, Riahi K, Govindasamy B, Bauer N, Eyring V, Fall CMN, Frieler K, Gidden MJ, Gohar LK, Högner A, Jones AD, Kikstra J, King A, Knutti R, Kriegler E, Lawrence P, Lennard C, Lowe J, Mathison C, Mehmood S, Nicholls Z, Prado LF, Zhang Q, Rose SK, Ruane AC, Sandstad M, Schleussner C-F, Seferian R, Sillmann J, **Smith C**, Sörensson AA, Panickal S, Tachiiri K, Vaughan N, Vishwanathan SS, Yokohata T, Zecchetto M, Ziehn T, 2026. The Scenario Model Intercomparison Project for CMIP7 (ScenarioMIP-CMIP7). Geoscientific Model Development, 19(7), 2627–2656. doi:10.5194/gmd-19-2627-2026
114. Xing J, Yuan J, **Smith C**, Li D, 2026. Multi-Century Projections of Dynamic Sea Level Based on an Updated Two-Layer Emulator. Geophysical Research Letters, 53(6), e2025GL120855. doi:10.1029/2025GL120855
113. Beath H, **Smith C**, Kikstra JS, Dekker MM, Gidden MJ, Rogelj J, 2026. A weighting framework to improve the use of emissions scenario ensembles of opportunity. Nature Climate Change. doi:10.1038/s41558-026-02565-5
112. Wells CD, Ramme L, **Smith C**, Breier J, Muralidhar A, Li C, Gjermundsen A, Schoenberg WA, Blanz B, Mauritzen C, 2026. FRIDA-Clim v1.0.1: a simple climate model with process-based carbon cycle used in the integrated assessment model FRIDAv2.1. Geoscientific Model Development, 19(4), 1429–1453. doi:10.5194/gmd-19-1429-2026
111. Wells CD, Blanz B, Ramme L, Breier J, Callegari B, Muralidhar A, Rajah JK, Lindqvist AN, Eriksson AE, Schoenberg WA, Köberle AC, Wang-Erlandsson L, Mauritzen C, Grimeland MB, **Smith C**, 2026. The representation of climate impacts in the FRIDAv2.1 Integrated Assessment Model. Geoscientific Model Development, 19(3), 1229–1260. doi:10.5194/gmd-19-1229-2026

110. Im U, Samset BH, Nenes A, Thomas JL, Kokkola H, Dubovik O, Amiridis V, Arola A, Bellouin N, Benedetti A, Bilde M, Blichner S, Decesari S, Ekman AML, García-Pando CP, Gross S, Gryspeerdt E, Hasekamp O, Kahn RA, Laakso A, Lohmann U, Marelle L, Massling AH, Myhre CL, Pöhlker M, Quaas J, Raatikainen T, Riipinen I, Schmale J, Seifert P, Skov H, **Smith C**, Sporre MK, Stier P, Storelvmo T, Tsigaridis K, Van Diedenhoven B, Virtanen A, Wandinger U, Wilcox LJ, Zieger P, 2026. Aerosol-Cloud Interactions: Overcoming a Barrier to Projecting Near-Term Climate Evolution and Risk. AGU Advances, 7(1), e2025AV001872. doi:10.1029/2025AV001872
109. Wells CD, Cummins DP, He H, **Smith C**, 2026. Long run emulator calibration increases warming and sea-level rise projections. Environmental Research Letters, 21(3), 034008. doi:10.1088/1748-9326/ae3847
108. Romero-Prieto A, Mathison C, **Smith C**, 2026. Review of climate simulation by Simple Climate Models. Geoscientific Model Development, 19(1), 115–165. doi:10.5194/gmd-19-115-2026

2025

107. Ramme L, Blanz B, Wells C, Wong TE, Schoenberg W, **Smith C**, Li C, 2025. Feedback-based sea level rise impact modelling for integrated assessment models with FRISIAv1.0. Geoscientific Model Development, 18(24), 10017–10052. doi:10.5194/gmd-18-10017-2025
106. Schoenberg W, Blanz B, Rajah JK, Callegari B, Wells C, Breier J, Grimeland MB, Lindqvist AN, Ramme L, **Smith C**, Li C, Mashhadi S, Muralidhar A, Mauritzen C, 2025. An overview of FRIDA v2.1: a feedback-based, fully coupled, global integrated assessment model of climate and humans. Geoscientific Model Development, 18(21), 8047–8069. doi:10.5194/gmd-18-8047-2025
105. Nauels A, Nicholls Z, Möller T, Hermans THJ, Mengel M, Kloenne U, **Smith C**, Slangen ABA, Palmer MD, 2025. Multi-century global and regional sea-level rise commitments from cumulative greenhouse gas emissions in the coming decades. Nature Climate Change. doi:10.1038/s41558-025-02452-5
104. Verkerk M, Aubry TJ, **Smith C**, Hopcroft PO, Sigl M, Tierney JE, Anchukaitis K, Osman M, Schmidt A, Toohey M, 2025. Using reduced-complexity volcanic aerosol and climate models to produce large ensemble simulations of Holocene temperature. Climate of the Past, 21(10), 1755–1778. doi:10.5194/cp-21-1755-2025
103. Sanderson BM, Brovkin V, Fisher RA, Hohn D, Ilyina T, Jones CD, Koenigk T, Koven C, Li H, Lawrence DM, Lawrence P, Liddicoat S, MacDougall AH, Mengis N, Nicholls Z, O'Rourke E, Romanou A, Sandstad M, Schwinger J, Séférian R, Sentman LT, Simpson IR, **Smith C**, Steinert NJ, Swann ALS, Tjiputra J, Ziehn T, 2025. flat10MIP: an emissions-driven experiment to diagnose the climate response to positive, zero and negative CO2 emissions. Geoscientific Model Development, 18(17), 5699–5724. doi:10.5194/gmd-18-5699-2025
102. Griffiths PT, Wilcox LJ, Allen RJ, Naik V, O'Connor FM, Prather M, Archibald A, Brown F, Deushi M, Collins W, Fiedler S, Oshima N, Murray LT, Samset BH, **Smith C**, Turnock S, Watson-Parris D, Young PJ, 2025. Opinion: The role of AerChemMIP in advancing climate and air quality research. Atmospheric Chemistry and Physics, 25(14), 8289–8328. doi:10.5194/acp-25-8289-2025
101. Malagón-Santos V, **Smith C**, Fredriksen H-B, Hermans THJ, Edwards TL, Slangen ABA, 2025. Emulating long-term CMIP6 projections of sterodynamic sea-level change using a three-layer energy balance model. Environmental Research Letters, 20(8), 084034. doi:10.1088/1748-9326/ade906
100. Myhre G, Samset BH, Stjern CW, Hodnebrog Ø, Kramer R, **Smith C**, Andrews T, Boucher O, Faluvegi G, Forster PM, Iversen T, Kirkevåg A, Olivié D, Shindell D, Stier P, Watson-Parris D, 2025. The warming effect of black carbon must be reassessed in light of observational constraints. Cell Reports Sustainability, 100428. doi:10.1016/j.crsus.2025.100428
99. Forster PM, **Smith C**, Walsh T, Lamb WF, Lamboll R, Cassou C, Hauser M, Hausfather Z, Lee J-Y, Palmer MD, Von Schuckmann K, Slangen ABA, Szopa S, Trewin B, Yun J, Gillett NP, Jenkins S, Matthews HD, Raghavan K, Ribes A, Rogelj J, Rosen D, Zhang X, Allen M, Aleluia Reis L, Andrew RM, Betts RA, Borger A, Broersma JA, Burgess SN, Cheng L, Friedlingstein P, Domingues CM, Gambarini M, Gasser T, Gütschow J, Ishii M, Kadow C, Kennedy J, Killick RE, Krummel PB, Liné A, Monselesan DP, Morice C, Mühle J, Naik V, Peters GP, Pirani A, Pongratz J, Minx JC, Rigby M, Rohde R, Savita A, Seneviratne SI, Thorne P, Wells C, Western LM, Van Der Werf GR, Wijffels SE, Masson-Delmotte V, Zhai

P, 2025. Indicators of Global Climate Change 2024: annual update of key indicators of the state of the climate system and human influence. Earth System Science Data, 17(6), 2641–2680. doi:10.5194/essd-17-2641-2025

98. Munday G, Jones CD, Steinert NJ, Mathison C, Burke EJ, **Smith C**, Huntingford C, Varney RM, Wiltshire AJ, 2025. Risks of unavoidable impacts on forests at 1.5 °C with and without overshoot. Nature Climate Change. doi:10.1038/s41558-025-02327-9
97. Chim MM, Aubry TJ, **Smith C**, Schmidt A, 2025. Neglecting future sporadic volcanic eruptions underestimates climate uncertainty. Communications Earth & Environment, 6(1), 236. doi:10.1038/s43247-025-02208-1
96. Pelz S, Ganti G, Lamboll R, Grant L, **Smith C**, Pachauri S, Rogelj J, Riahi K, Thiery W, Gidden MJ, 2025. Using net-zero carbon debt to track climate overshoot responsibility. Proceedings of the National Academy of Sciences, 122(13), e2409316122. doi:10.1073/pnas.2409316122
95. Mathison C, Burke EJ, Munday G, Jones CD, **Smith CJ**, Steinert NJ, Wiltshire AJ, Huntingford C, Kovacs E, Gohar LK, Varney RM, McNeall D, 2025. A rapid-application emissions-to-impacts tool for scenario assessment: Probabilistic Regional Impacts from Model patterns and Emissions (PRIME). Geoscientific Model Development, 18(5), 1785–1808. doi:10.5194/gmd-18-1785-2025
94. Storelvmo T, Yuan M, Leirvik T, Alterskjær K, Phillips PCB, **Smith C**, 2025. Assessing the robustness and implications of econometric estimates of climate sensitivity. Environmental Research Letters, 20(2), 024055. doi:10.1088/1748-9326/adabfc
93. Pfleiderer P, Frölicher TL, Kropf CM, Lamboll RD, Lejeune Q, Capela Lourenço T, Maussion F, McCaughey JW, Quilcaille Y, Rogelj J, Sanderson B, Schuster L, Sillmann J, **Smith C**, Theokritoff E, Schleussner C-F, 2025. Reversal of the impact chain for actionable climate information. Nature Geoscience. doi:10.1038/s41561-024-01597-w
92. Tsutsui J, **Smith C**, 2025. Revisiting two-layer energy balance models for climate assessment. Environmental Research Letters, 20(1), 014059. doi:10.1088/1748-9326/ad9ec5

2024

91. **Smith C**, Cummins DP, Fredriksen H-B, Nicholls Z, Meinshausen M, Allen M, Jenkins S, Leach N, Mathison C, Partanen A-I, 2024. fair-calibrate v1.4.1: calibration, constraining, and validation of the FaIR simple climate model for reliable future climate projections. Geoscientific Model Development, 17(23), 8569–8592. doi:10.5194/gmd-17-8569-2024
90. Friedlingstein P, Artaxo P, Gallego-Sala AV, Jia G, Jones CD, Kawamiya M, Loisel J, Loutre M-F, Rehfeld K, Rovere A, **Smith C**, Séférian R, Van Der Wel N, Ziegler E, 2024. Earth system responses to different levels of greenhouse gas emissions mitigation. Frontiers in Climate, 6, 1480208. doi:10.3389/fclim.2024.1480208
89. Sanderson BM, Booth BBB, Dunne J, Eyring V, Fisher RA, Friedlingstein P, Gidden MJ, Hajima T, Jones CD, Jones CG, King A, Koven CD, Lawrence DM, Lowe J, Mengis N, Peters GP, Rogelj J, **Smith C**, Snyder AC, Simpson IR, Swann ALS, Tebaldi C, Ilyina T, Schleussner C-F, Séférian R, Samset BH, van Vuuren D, Zaehle S, 2024. The need for carbon-emissions-driven climate projections in CMIP7. Geoscientific Model Development, 17(22), 8141–8172. doi:10.5194/gmd-17-8141-2024
88. Quaas J, Andrews T, Bellouin N, Block K, Boucher O, Ceppi P, Dagan G, Doktorowski S, Eichholz HM, Forster P, Goren T, Gryspeerdt E, Hodnebrog Ø, Jia H, Kramer R, Lange C, Maycock AC, Mülmenstädt J, Myhre G, O'Connor FM, Pincus R, Samset BH, Senf F, Shine KP, **Smith C**, Stjern CW, Takemura T, Toll V, Wall CJ, 2024. Adjustments to Climate Perturbations—Mechanisms, Implications, Observational Constraints. AGU Advances, 5(5), e2023AV001144. doi:10.1029/2023AV001144
87. Schleussner C-F, Ganti G, Lejeune Q, Zhu B, Pfleiderer P, Prütz R, Ciais P, Frölicher TL, Fuss S, Gasser T, Gidden MJ, Kropf CM, Lacroix F, Lamboll R, Martyr R, Maussion F, McCaughey JW, Meinshausen M, Mengel M, Nicholls Z, Quilcaille Y, Sanderson B, Seneviratne SI, Sillmann J, **Smith CJ**, Steinert NJ, Theokritoff E, Warren R, Price J, Rogelj J, 2024. Overconfidence in climate overshoot. Nature, 634(8033), 366–373. doi:10.1038/s41586-024-08020-9
86. Allen RJ, Zhao X, Randles CA, Kramer RJ, Samset BH, **Smith CJ**, 2024. Present-day methane shortwave absorption mutes surface warming relative to preindustrial conditions. Atmospheric Chemistry and Physics,

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